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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Computer Science & Engg.) (CBCS) Degree Examinations

May - June 2024

23CSE201 – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Given 2 jugs, a 5- gallon (5-g) and the other 3-gallon (3-g) with no measuring marker on them. There is endless supply of water through tap. Task is to get 4-gallon of water in the 5-g jug				
	• Write down the production rules for the above problem	10	L*3	1	1
	• Write any one solution to the above problem				
b)	Explain is AI a science or is it engineering? Or neither or both?	10	L2	1	1
2. a)	What are the components of well-defined problems? How will you measure the problem-solving performance?	10	L2	1	1
b)	What is uninformed search and explain DFS and BFS with example.	10	L1	1	1
	Unit – II				
3. a)	Describe the Knowledge Representation and mappings in AI	10	L2	2	1
b)	Illustrate the syntax of Propositional logic in detail.	10	L2	2	1
4. a)	Illustrate the types of Inference Rules	10	L2	2	1
b)	Explain how Predicate logic helps in knowledge representation in AI	10	L2	2	1
	Unit – III				
5. a)	Illustrate the situation calculus rule for the sentence "Get a quart of milk, a bunch of bananas and a variable-speed cordless drill"	10	L2	3	1
b)	Outline the complete partial Order planning algorithm.	10	L2	3	1
6. a)	What are the basic components of situation calculus? How does situation calculus differ from other AI formalism?	10	L1	3	1
b)	Illustrate the Partial order planning of Wearing Shoe	10	L2	3	1
	Unit – IV				
7. a)	Compare supervised learning and unsupervised learning. List the algorithms for each type of learning	5	L3	4	1
b)	Write Baye's theorem and elaborate the meaning of each component	5	L1	4	1
c)	Explain the concepts of clustering approaches. How it differ from classification	5	L2	4	1
d)	List the applications of clustering and identify advantages and disadvantages of clustering algorithm	5	L1	4	1
8. a)	State the Baye's theorem. How it is useful for decision making under uncertainty?	10	L1	4	1
b)	Explain in detail about k-means algorithm?	10	L2	4	1
	Unit – V				
9. a)	Explain SVM Algorithm in detail.	10	L2	5	1
b)	Explain CADET System using Case based reasoning.	10	L2	5	1
10. a)	Explain the classifications of SVM in detail	10	L2	5	1
b)	List the generic properties of case-based reasoning system. Differentiate lazy learning and eager learning	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
